

§11.11 The Binomial Series

Homework : 3,7,11,15,17

牛頓推廣 binomial 的展開，即 $(a+b)^k$ ， $k \in N$ ，的展開，推廣至 $(a+b)^k$ ， $k \in R$ 。

Theorem1 : If $k \in R$, $|x| < 1$, then $(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n$,

$$\binom{k}{n} = \frac{(k)(k-1)\dots(k-n+1)}{n!} (n \geq 1) \text{ and } \binom{k}{0} = 1.$$

Remark : 目前我們學到 3 個方法得 Taylor series or power series representation.

$$(i) \frac{1}{1-x} \quad (ii) \ c_n = \frac{f^n(a)}{n!} \quad (iii) \text{ 多項式展開.}$$

Example 1 :

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{4-x}} \\ &= \frac{1}{2} \left(1 - \frac{x}{4}\right)^{-\frac{1}{2}} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \left(-\frac{x}{4}\right)^n \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n 1.3 \dots (2n-1)}{2^n n!} (-1)^n \frac{x^n}{2^{2n}} \\ &= \sum_{n=0}^{\infty} \frac{1.3 \dots (2n-1)}{2^{3n+1} n!} x^n \\ &\left(\binom{-\frac{1}{2}}{n} = \frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\dots\left(-\frac{1}{2}-n+1\right)}{n!} \right) \end{aligned}$$

Example 2 : Let $f(x) = (1+x^2)^{\frac{1}{2}}$. Find $f^{(10)}(0)$.

Solution :

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} (x^2)^n \quad (\text{多項式展開}) \\ &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k \quad (\text{Maclaurin Series}). \end{aligned}$$

$$\Rightarrow k=10 \Rightarrow n=5.$$

$$\Rightarrow \frac{f^{(10)}(0)}{10!} = \binom{\frac{1}{2}}{5} = \frac{\frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \left(-\frac{5}{2}\right) \left(-\frac{7}{2}\right)}{5!} = \frac{7}{2^8}$$

$$\Rightarrow f^{(10)}(0) = \frac{7}{2^8} 10!$$

Example 3 : Let $f(x) = (1+x^3)^{\frac{1}{2}}$. Find $f^{(9)}(0)$.

Solution :

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} (x^3)^n \\ &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k \end{aligned}$$

$$\Rightarrow k=9 \Rightarrow n=3.$$

$$\Rightarrow \frac{f^{(9)}(0)}{9!} = \binom{\frac{1}{2}}{3} = \frac{\left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \left(-\frac{5}{2}\right)}{3!} = -\frac{5}{2^4}$$

$$\Rightarrow f^{(9)}(0) = \frac{5}{2^4} 9!$$